

Digital Logic Design Report: 4-Way Traffic Control System

Digital Logic Design Team

November 14, 2025

1 Introduction and Objectives

The objective of this project is to design a robust traffic control system using digital logic principles. The system consists of two major control structures:

- 1. Minor Cycle (0–5 Light Sequence):** A six-state finite state machine (FSM) that controls Red, Red+Yellow, three phases of Green, and Yellow.
- 2. Major Cycle (Post Rotation):** A 2-bit synchronous counter that selects one of the four traffic posts, changing only when the Minor Cycle completes.

The system is suitable for hardware implementation or simulation using tools such as Proteus ISIS.

2 System Architecture

The system follows a hierarchical Master–Slave structure. The Master Controller manages post rotation, while each Slave Unit manages a full 6-state light sequence.

Table 1: System Component Architecture

Component	Function	Inputs/Outputs
0–5 Counter	Stores the 3-bit state (A, B, C) for the light sequence.	Master Clock, trigger x , outputs A, B, C .
2-Bit Counter	Rotates between posts (00–11).	Trigger Clock.
2:4 Decoder	Selects one of four posts.	Q_1, Q_0 inputs; T_1 – T_4 outputs.

3 Sequence Generator Logic (0–5 Counter)

The Minor Cycle consists of a six-state FSM implemented using three JK Flip-Flops storing bits (A, B, C).

3.1 State Definitions

Table 2: Traffic Light State Definitions

ABC	Decimal	Light State	Transition Trigger
000	0	Red	Wait; proceeds to 001 only if $x = 1$.
001	1	Red + Yellow	Preparing for Green.
010	2	Green (Phase 1)	Traffic flows.
011	3	Green (Phase 2)	Sustains Green.
100	4	Green (Phase 3)	Sustains Green.
101	5	Yellow	Ends cycle; activates post change.

3.2 JK Excitation Equations

$$\begin{aligned} J_A &= BC, & K_A &= 1, \\ J_B &= \overline{C} + \overline{A}, & K_B &= 1, \\ J_C &= (x\overline{A}\overline{B}) + (A\overline{B}), & K_C &= \overline{B}. \end{aligned}$$

4 Light Decoding Logic

$$R = \overline{A}\overline{B}$$

$$Y = \overline{B}C$$

$$G = \overline{A}B + A\overline{B}\overline{C}$$

5 Master Control and Synchronization

5.1 Next Post Trigger Logic

$$\text{CLK}_{2\text{-bit}} = (A\overline{B}C) \cdot \text{CLK}_{\text{Master}}$$

5.2 Post Selection Logic

Table 3: 2-Bit Counter Excitation

Flip-Flop	J Input	K Input
Q_0 (LSB)	$J_0 = T$	$K_0 = T$
Q_1 (MSB)	$J_1 = TQ_0$	$K_1 = TQ_0$

The outputs (Q_1, Q_0) feed the 2:4 Decoder, activating one of T_1 - T_4 .

Appendix: Circuit Diagrams

Below are the circuit diagrams used in the design and simulation of the traffic control system.

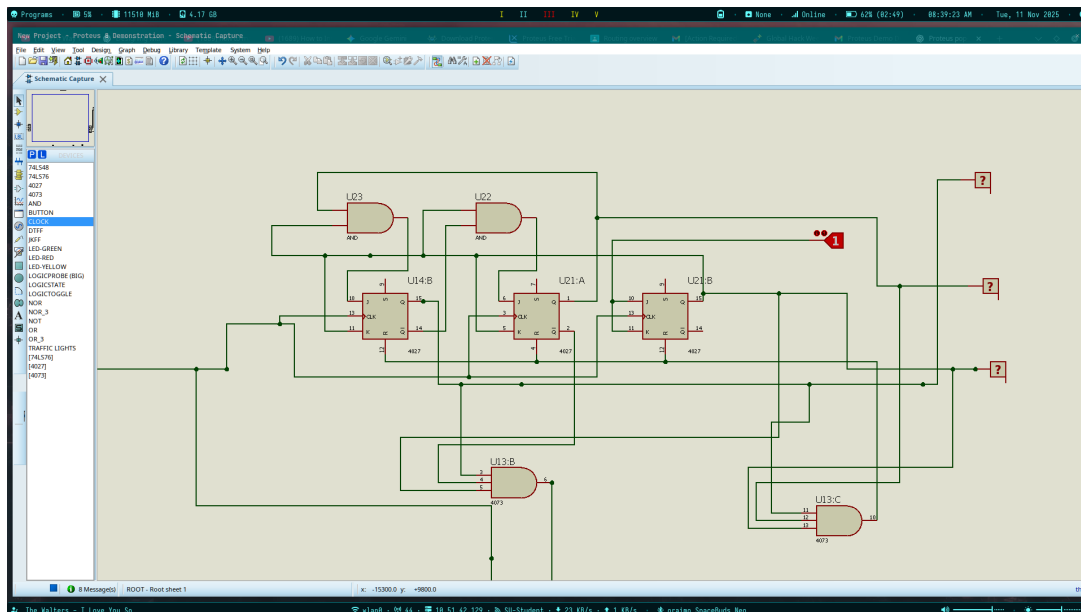


Figure 1: 0-5 State Machine for the Light Sequence

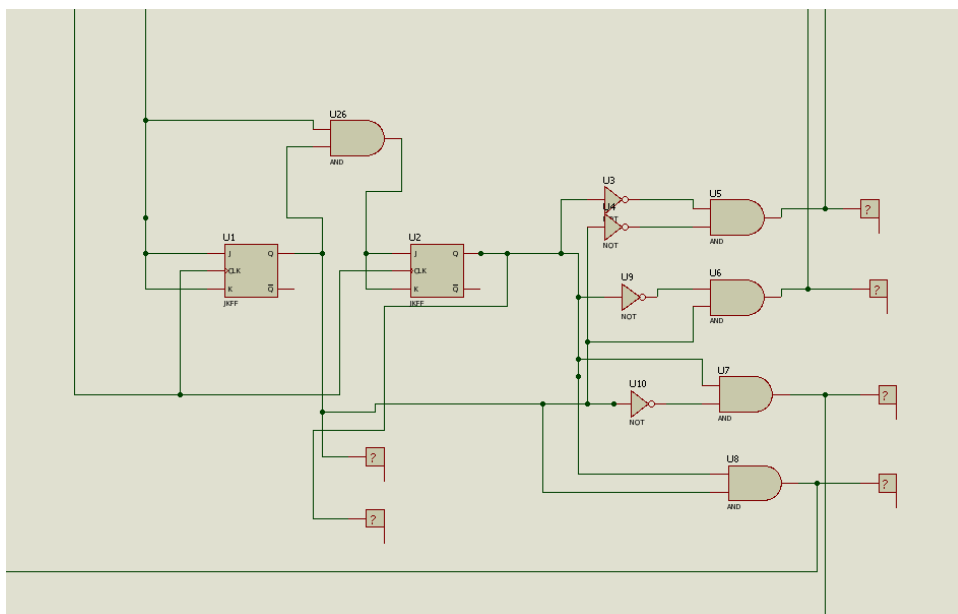


Figure 2: 2 Bit Counter and 2:4 Decoder for Post Selection

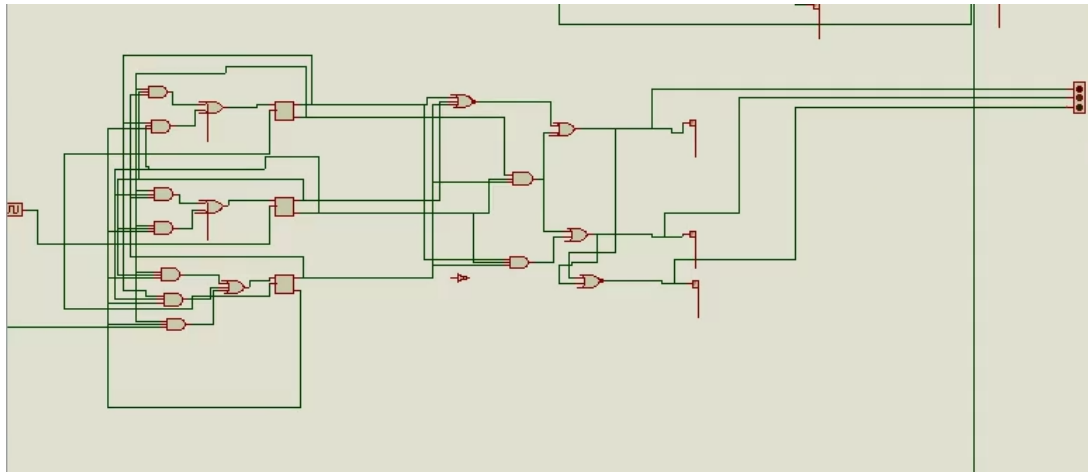


Figure 3: Four-Post Traffic System Layout